

AMAN PARGANIHA

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EDUCATION

International Institute of Information Technology, Naya Raipur — India

Aug 2025 – Aug 2027

Master of Technology in Computer Science & Engineering (AI/ML) — CGPA: 8.09 / 10.0

- Relevant Coursework: Advanced Algorithms, Software Systems, GenAI, Cloud Computing, Cybersecurity

Shri Shankaracharya Technical Campus, Bhilai — India

Aug 2019 – Aug 2023

Bachelor of Technology in Computer Science & Engineering — CGPA: 8.22 / 10.0

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

TECHNICAL SKILLS

Programming Languages: C++, Python, SQL

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, NLTK, NLP, Deep Learning, Feature Engineering, Model Evaluation

Data Engineering & Tools: Pandas, NumPy, Git, Linux, AWS, Data Pipelines, ETL, Version Control

Core Computer Science: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Software Architecture: Scalable Systems Design, Modular Architecture, Design Patterns, Code Optimization

Software Engineering Practices: Clean Code, Modular Design, Unit Testing, Debugging, Code Reviews

PROJECTS

Event Driven Retrieval-Augmented Generation Agent [Code](#)

Python, FastAPI, Inngest, Qdrant, OpenAI API, Docker

- Architected an end-to-end RAG system that asynchronously processes PDFs using Inngest workflows to handle large-scale document ingestion without blocking the UI.
- Developed a semantic retrieval engine using Qdrant and LlamaIndex, optimizing chunk sizes and embedding strategies to maximize retrieval precision.
- Integrated OpenAI's GPT-4o with context-aware prompting to generate accurate answers from indexed technical documents.
- Deployed a full-stack solution with a FastAPI backend and Streamlit frontend, containerized via Docker for reproducible deployment.

Multimodal Credit Risk Analysis using SEC XBRL & NLP [Code](#)

Python, Pandas, Scikit-learn, NLP, Git

- Processed SEC XBRL financial filings (2022–2024) and engineered financial ratios for corporate credit risk analysis.
- Combined structured financial data with NLP-derived features (sentiment, risk, uncertainty) to build a multimodal dataset.
- Trained and evaluated binary and multiclass credit rating models using standard machine-learning algorithms.

TECHNICAL EXPERIENCE & CERTIFICATIONS

5-Day AI Agents Intensive Course (Google × Kaggle) | [Certification](#)

- Completed an intensive program on agentic workflows, orchestration, and task automation using modern AI techniques.

Walmart USA – Advanced Software Engineering Job Simulation | [Certification](#)

- Completed an industry-oriented simulation focused on data structures, algorithms, and scalable system design.

ACHIEVEMENTS & STRENGTHS

- Strong CS Fundamentals: OS, Computer Networks, DBMS, OOP, and Software Architecture.
- Reproducible ML/NLP Pipelines: Built scalable ML/NLP pipelines with emphasis on maintainability and best practices.
- Problem-Solving Excellence: 200+ LeetCode problems solved; consistently top-ranked in programming coursework.
- GATE CSE: Ranked among the Top 7.8% nationwide out of approximately 2.1 lakh candidates, securing admission to M.Tech (AI/ML) at IIIT Naya Raipur.
- Open Source — Matplotlib: Merged [PR #31422](#) into main (v3.11.0), improving legend parameter documentation (issue #26620); reviewed and approved by core maintainer timhoffm. Submitted a GSoC 2026 proposal to NumFOCUS for the *Indirect Transforms: Reusable Positioning API* project.
- Open Source — JupyterLab: Approved [PR #18668](#) improving user-facing documentation through multi-round review with core maintainers. Contributed [PR #18610](#) adding Playwright end-to-end test coverage.